```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

print("Libraries Imported Successfully")
```

```
Libraries Imported Successfully
```

Dataset Upload

```
from google.colab import files
uploaded = files.upload()
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving WA_Fn-UseC_-HR-Employee-Attrition.csv to WA_Fn-UseC_-HR-Employee-At
```

```
df = pd.read_csv("WA_Fn-UseC_-HR-Employee-Attrition.csv")
df.head()
```

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome
0	41	Yes	Travel_Rarely	1102	Sales	1
1	49	No	Travel_Frequently	279	Research & Development	8
2	37	Yes	Travel_Rarely	1373	Research & Development	2
3	33	No	Travel_Frequently	1392	Research & Development	3
4	27	No	Travel_Rarely	591	Research & Development	2

5 rows × 35 columns

```
print("Rows and Columns:")
print(df.shape)

print("\nColumn Names:")
print(df.columns.tolist())
```

```
Rows and Columns:
(1470, 35)

Column Names:
```

```
['Age', 'Attrition', 'BusinessTravel', 'DailyRate', 'Department', 'Distanc
```

Dataset Overview

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 35 columns):
 #   Column                    Non-Null Count  Dtype 
---  ------                    --------------  ----- 
 0   Age                       1470 non-null   int64 
 1   Attrition                 1470 non-null   object
 2   BusinessTravel            1470 non-null   object
 3   DailyRate                 1470 non-null   int64 
 4   Department                1470 non-null   object
 5   DistanceFromHome          1470 non-null   int64 
 6   Education                 1470 non-null   int64 
 7   EducationField            1470 non-null   object
 8   EmployeeCount             1470 non-null   int64 
 9   EmployeeNumber            1470 non-null   int64 
 10  EnvironmentSatisfaction   1470 non-null   int64 
 11  Gender                    1470 non-null   object
 12  HourlyRate                1470 non-null   int64 
 13  JobInvolvement            1470 non-null   int64 
 14  JobLevel                  1470 non-null   int64 
 15  JobRole                   1470 non-null   object
 16  JobSatisfaction           1470 non-null   int64 
 17  MaritalStatus             1470 non-null   object
 18  MonthlyIncome             1470 non-null   int64 
 19  MonthlyRate               1470 non-null   int64 
 20  NumCompaniesWorked        1470 non-null   int64 
 21  Over18                    1470 non-null   object
 22  OverTime                  1470 non-null   object
 23  PercentSalaryHike         1470 non-null   int64 
 24  PerformanceRating         1470 non-null   int64 
 25  RelationshipSatisfaction  1470 non-null   int64 
 26  StandardHours             1470 non-null   int64 
 27  StockOptionLevel          1470 non-null   int64 
 28  TotalWorkingYears         1470 non-null   int64 
 29  TrainingTimesLastYear     1470 non-null   int64 
 30  WorkLifeBalance           1470 non-null   int64 
 31  YearsAtCompany            1470 non-null   int64 
 32  YearsInCurrentRole        1470 non-null   int64 
 33  YearsSinceLastPromotion   1470 non-null   int64 
 34  YearsWithCurrManager      1470 non-null   int64 
dtypes: int64(26), object(9)
memory usage: 402.1+ KB
```

```
df.isnull().sum()
```

	0
Age	0
Attrition	0
BusinessTravel	0
DailyRate	0
Department	0
DistanceFromHome	0
Education	0
EducationField	0
EmployeeCount	0
EmployeeNumber	0
EnvironmentSatisfaction	0
Gender	0
HourlyRate	0
JobInvolvement	0
JobLevel	0
JobRole	0
JobSatisfaction	0
MaritalStatus	0
MonthlyIncome	0
MonthlyRate	0
NumCompaniesWorked	0
Over18	0
OverTime	0
PercentSalaryHike	0
PerformanceRating	0
RelationshipSatisfaction	0
StandardHours	0
StockOptionLevel	0
TotalWorkingYears	0
TrainingTimesLastYear	0
WorkLifeBalance	0

```
df.duplicated().sum()
```

YearsInCurrentRole	0
YearsSinceLastPromotion	0

dtype: int64

```
np.int64(0)
```

```
df.describe()
```

	Age	DailyRate	DistanceFromHome	Education	EmployeeCoun
count	1470.000000	1470.000000	1470.000000	1470.000000	1470.
mean	36.923810	802.485714	9.192517	2.912925	1.
std	9.135373	403.509100	8.106864	1.024165	0.
min	18.000000	102.000000	1.000000	1.000000	1.
25%	30.000000	465.000000	2.000000	2.000000	1.
50%	36.000000	802.000000	7.000000	3.000000	1.
75%	43.000000	1157.000000	14.000000	4.000000	1.
max	60.000000	1499.000000	29.000000	5.000000	1.

8 rows × 26 columns

Data Cleaning

```
df['Attrition'].value_counts()
```

	count
Attrition	
No	1233
Yes	237

dtype: int64

```
df['Attrition'].value_counts(normalize=True)*100
```

	proportion
Attrition	
No	83.877551
Yes	16.122449

dtype: float64

```
df.isnull().sum()
```

	0
Age	0
Attrition	0
BusinessTravel	0
DailyRate	0
Department	0
DistanceFromHome	0
Education	0
EducationField	0
EmployeeCount	0
EmployeeNumber	0
EnvironmentSatisfaction	0
Gender	0
HourlyRate	0
JobInvolvement	0
JobLevel	0
JobRole	0
JobSatisfaction	0
MaritalStatus	0
MonthlyIncome	0
MonthlyRate	0
NumCompaniesWorked	0
Over18	0
OverTime	0
PercentSalaryHike	0
PerformanceRating	0
RelationshipSatisfaction	0
StandardHours	0
StockOptionLevel	0
TotalWorkingYears	0
TrainingTimesLastYear	0
WorkLifeBalance	0

```
df.duplicated().sum()
```

```
np.int64(0)
```

YearsInCurrentRole 0

YearsSinceLastPromotion 0

```
print(df['EmployeeCount'].value_counts())
print(df['Over18'].value_counts())
print(df['StandardHours'].value_counts())
```

```
EmployeeCount
1    1470
Name: count, dtype: int64
Over18
Y    1470
Name: count, dtype: int64
StandardHours
80    1470
Name: count, dtype: int64
```

```
df.drop(
    columns=[
        'EmployeeCount',
        'Over18',
        'StandardHours'
    ],
    inplace=True
)

df.shape
```

```
(1470, 32)
```

```
df['AgeGroup'] = pd.cut(
    df['Age'],
    bins=[18,25,35,45,60],
    labels=['18-25','26-35','36-45','46+']
)

df[['Age','AgeGroup']].head()
```

	Age	AgeGroup
0	41	36-45
1	49	46+
2	37	36-45
3	33	26-35
4	27	26-35

```
df['AgeGroup'].value_counts()
```

AgeGroup	count
26-35	606
36-45	468
46+	273
18-25	115

dtype: int64

```
df['SalaryBand'] = pd.cut(
    df['MonthlyIncome'],
    bins=[0,5000,10000,15000,25000],
    labels=[
        'Low Income',
        'Medium Income',
        'High Income',
        'Very High Income'
    ]
)

df[['MonthlyIncome','SalaryBand']].head()
```

	MonthlyIncome	SalaryBand
0	5993	Medium Income
1	5130	Medium Income
2	2090	Low Income
3	2909	Low Income
4	3468	Low Income

```
df['SalaryBand'].value_counts()
```

SalaryBand	count
Low Income	749
Medium Income	440
High Income	148
Very High Income	133

dtype: int64

```
df.shape
```

```
(1470, 34)
```

```
df['AgeGroup'].value_counts()
```

AgeGroup	count
26-35	606
36-45	468
46+	273
18-25	115

dtype: int64

```
df['AgeGroup'] = pd.cut(
    df['Age'],
    bins=[17,25,35,45,60],
    labels=['18-25','26-35','36-45','46+']
)
```

```
df['AgeGroup'].value_counts()
```

AgeGroup	count
26-35	606
36-45	468
46+	273
18-25	123

dtype: int64

```
df['SalaryBand'].value_counts()
```

SalaryBand	count
Low Income	749
Medium Income	440
High Income	148
Very High Income	133

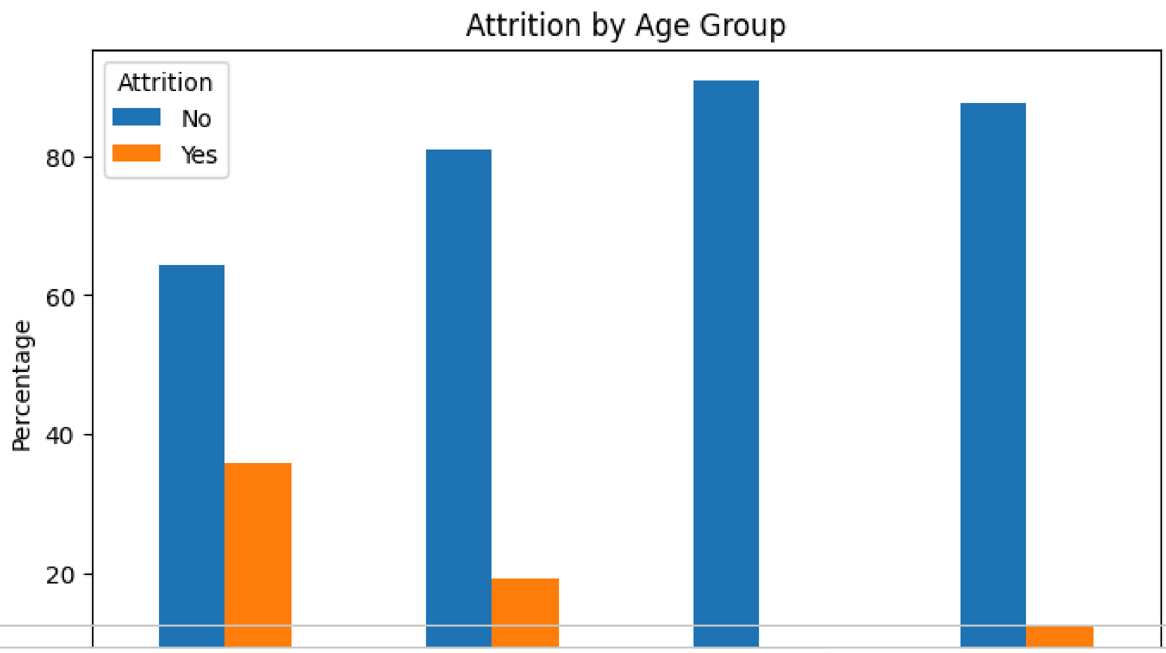
dtype: int64

Analysis 1: Age Group vs Attrition

```
age_attrition_pct = pd.crosstab(
    df['AgeGroup'],
    df['Attrition'],
    normalize='index'
)*100

round(age_attrition_pct,2)
```

Attrition / AgeGroup	No	Yes
18-25	64.23	35.77
26-35	80.86	19.14
36-45	90.81	9.19
46+	87.55	12.45

```
age_attrition_pct.plot(
    kind='bar',
    figsize=(8,5)
)

plt.title("Attrition by Age Group")
plt.ylabel("Percentage")
plt.show()
```




```
df['AgeGroup'].isnull().sum()
```

```
np.int64(0)
```

```
age_attrition_pct
```

Attrition	No	Yes
AgeGroup		
18-25	64.227642	35.772358
26-35	80.858086	19.141914
36-45	90.811966	9.188034
46+	87.545788	12.454212

- Observation

Employees aged 18–25 have the highest attrition rate at 35.77%, which is significantly higher than all other age groups.

Employees aged 36–45 show the lowest attrition rate at 9.19%, indicating higher stability and retention.

- Key Finding #1

Young employees (18–25) have highest attrition (35.8%).

- HR Recommendation

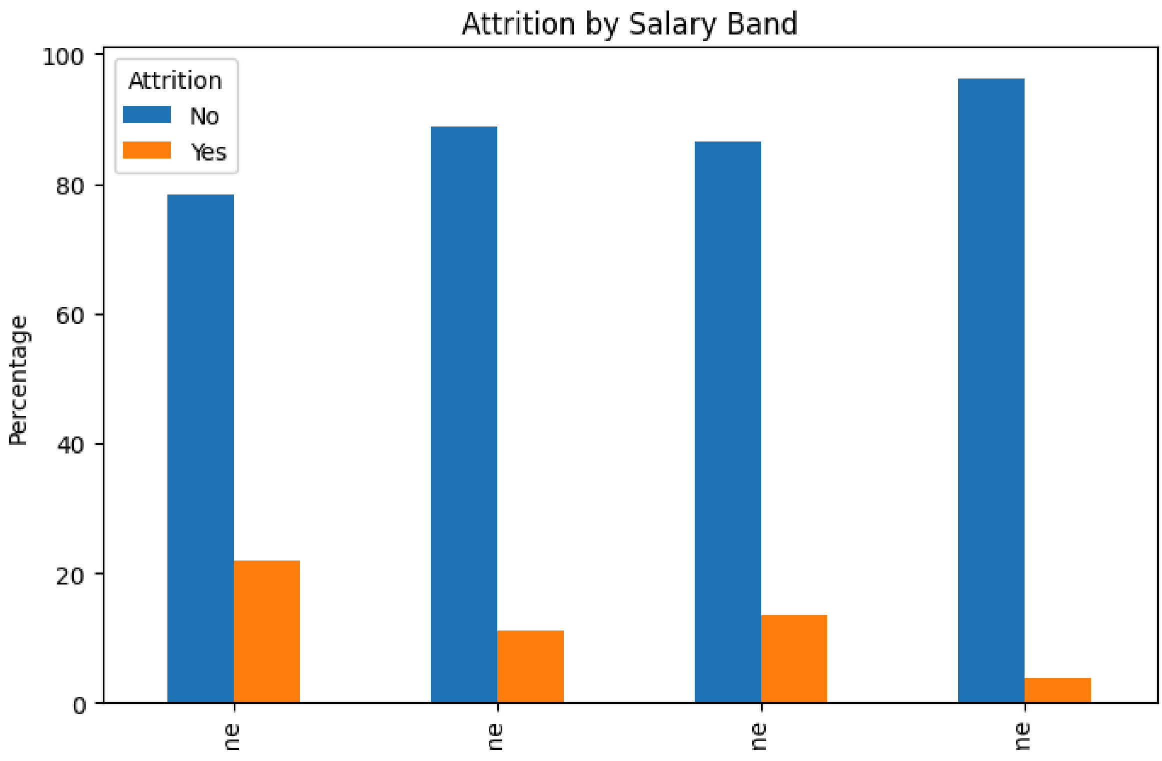
HR should focus on onboarding programs, career growth opportunities, mentorship, and engagement initiatives for younger employees to improve retention.

Analysis 2: Salary vs Attrition

```
salary_attrition_pct = pd.crosstab(
    df['SalaryBand'],
    df['Attrition'],
    normalize='index'
)*100

round(salary_attrition_pct,2)
```

Attrition	No	Yes
SalaryBand		
Low Income	78.24	21.76
Medium Income	88.86	11.14
High Income	86.49	13.51
Very High Income	96.24	3.76

```
salary_attrition_pct.plot(
    kind='bar',
    figsize=(8,5)
)

plt.title("Attrition by Salary Band")
plt.ylabel("Percentage")
plt.show()
```



- Observation

Employees in the Low Income category have the highest attrition rate (21.76%).

Employees in the Very High Income category have the lowest attrition rate (3.76%).

Attrition generally decreases as employee income increases.

- Key Finding #2

Low-income employees have highest attrition (21.8%).

- HR Recommendation

HR should review compensation structures, salary progression plans, and reward programs for lower-income employees to improve retention.

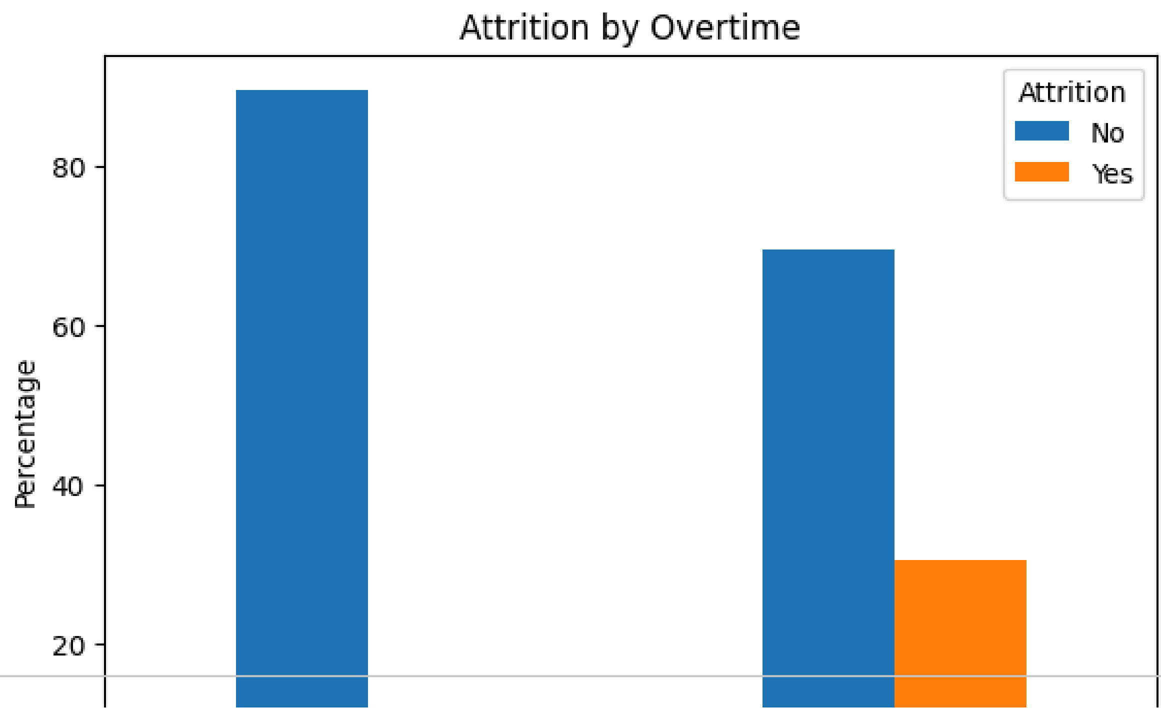
Analysis 3: Overtime vs Attrition

```
overtime_attrition_pct = pd.crosstab(
    df['OverTime'],
    df['Attrition'],
    normalize='index'
)*100

round(overtime_attrition_pct,2)
```

Attrition	No	Yes
OverTime		
No	89.56	10.44
Yes	69.47	30.53

```
overtime_attrition_pct.plot(
    kind='bar',
    figsize=(7,5)
)

plt.title("Attrition by Overtime")
```

```
plt.ylabel("Percentage")
plt.show()
```



- Observation

Employees working overtime have an attrition rate of 30.53%.

Employees not working overtime have an attrition rate of only 10.44%.

Overtime employees are approximately 3 times more likely to leave the organization.

- Key Finding #3

Overtime employees have highest attrition (30.5%).

- HR Recommendation

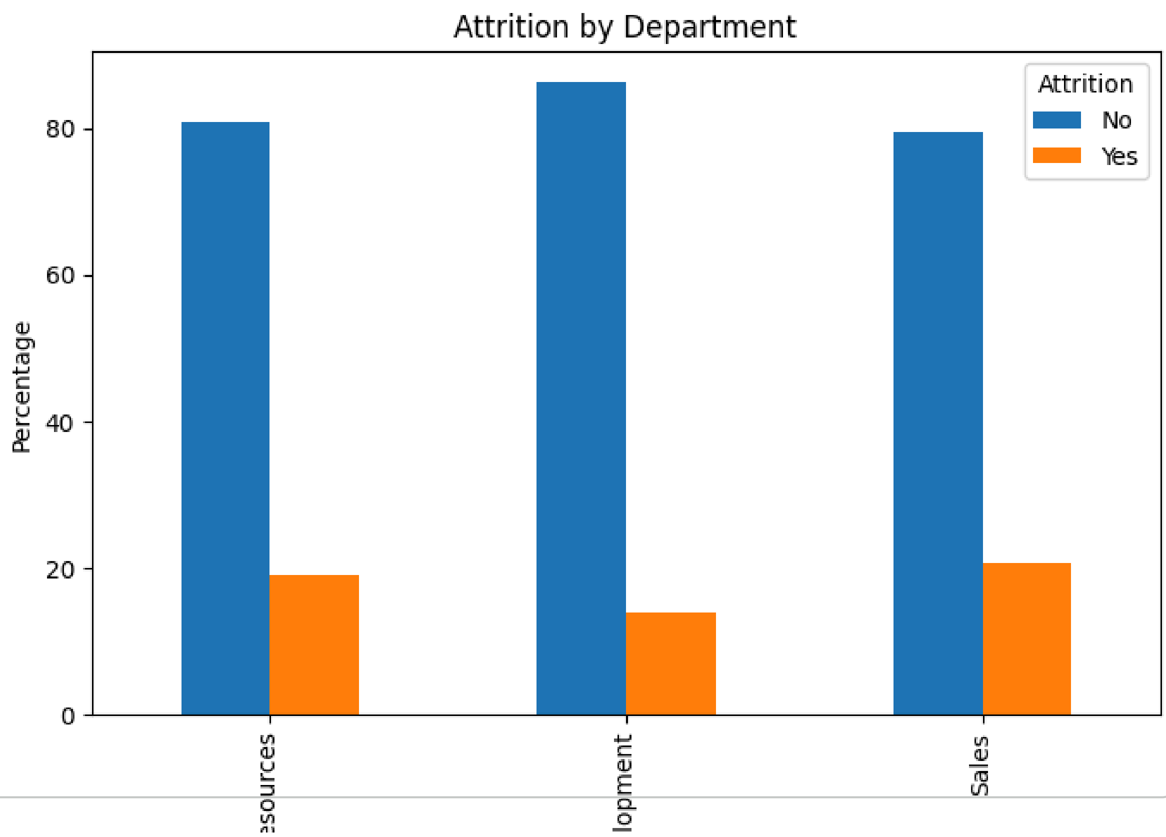
HR should monitor employee workload, reduce excessive overtime, and promote work-life balance initiatives to improve retention.

Analysis 4: Department vs Attrition

```
dept_attrition_pct = pd.crosstab(
    df['Department'],
    df['Attrition'],
    normalize='index'
)*100

round(dept_attrition_pct,2)
```

Attrition	No	Yes
Department		
Human Resources	80.95	19.05
Research & Development	86.16	13.84
Sales	79.37	20.63

```
dept_attrition_pct.plot(
    kind='bar',
    figsize=(8,5)
)

plt.title("Attrition by Department")
plt.ylabel("Percentage")
plt.show()
```



- Observation

Sales department has the highest attrition rate, followed closely by Human Resources.

- Key Finding #4

Sales department experiences the highest attrition

- HR Recommendation

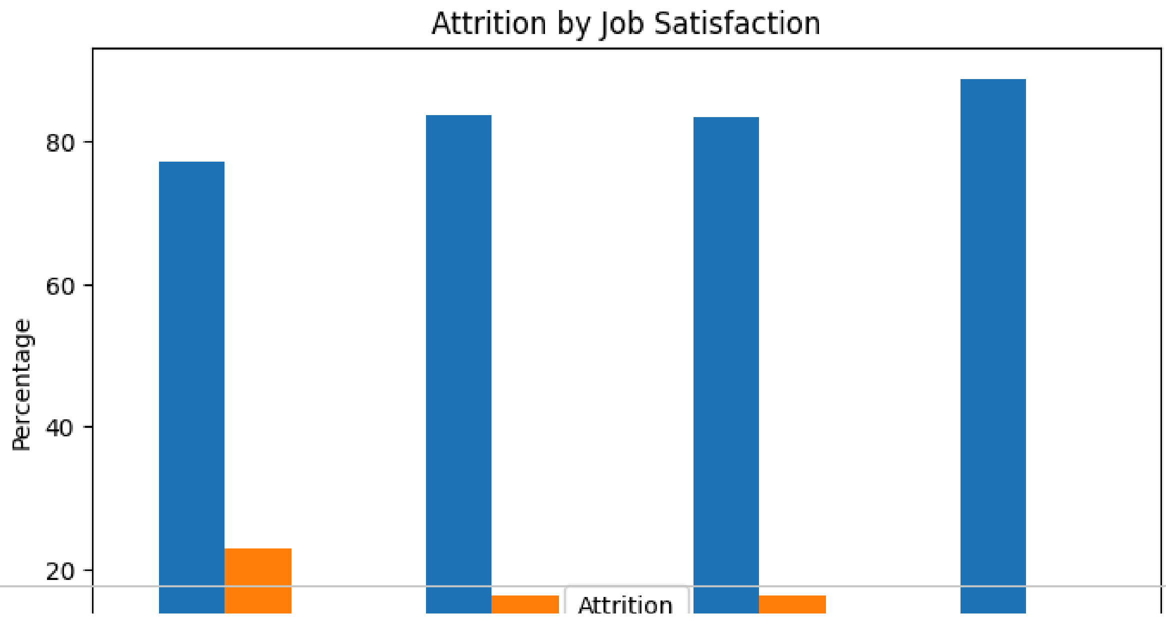
HR should conduct department-specific retention analysis in Sales to identify factors contributing to employee turnover.

Analysis 4: Job Satisfaction vs Attrition

```
jobsat_attrition = pd.crosstab(
    df['JobSatisfaction'],
    df['Attrition'],
    normalize='index'
) * 100

round(jobsat_attrition, 2)
```

Attrition	No	Yes
JobSatisfaction		
1	77.16	22.84
2	83.57	16.43
3	83.48	16.52
4	88.67	11.33

```
jobsat_attrition.plot(
    kind='bar',
    figsize=(8,5)
)

plt.title("Attrition by Job Satisfaction")
plt.ylabel("Percentage")
plt.show()
```



- Observation

Employees with the lowest job satisfaction (Level 1) have the highest attrition rate at 22.84%.

Employees with the highest job satisfaction (Level 4) have the lowest attrition rate at 11.33%.

Attrition generally decreases as job satisfaction increases.

- Key Finding #5

Low job satisfaction is strongly associated with attrition

- HR Recommendation

HR should focus on employee engagement, recognition programs, career development opportunities, and feedback mechanisms to improve job satisfaction and reduce turnover

```
df.to_csv(
    "HR_Analytics_Cleaned.csv",
    index=False
)
```